

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT:- ETHICAL HACKING

Practical no: 3

Aim: Linux Network Analysis and ARP Poisoning

- **PING:** A command used to check whether another computer or website is reachable over a network and to measure response time.
- **IPCONFIG:** A command used to display and manage the IP configuration of your computer, including the IP address, subnet mask, and default gateway.
- **NETSTAT:** A command used to display active network connections, listening ports, and network statistics.

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT:- ETHICAL HACKING

IPconfig

```
Microsoft Windows [Version 10.0.19040.865]
(c) Microsoft Corporation. All rights reserved.

C:\Users\AmLuc>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection* 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    IP Address . . . . . : 2005:205:5:88fc::add6:fa28:d3a5:a2e7
    Temporary IPv6 Address . . . . : 2405:205:5:88fc::2d9d:51b:8fd:51de
    Link-local IPv6 Address . . . . : fe80::8308::caaf:f960::ad89%5
    IPv4 Address. . . . . : 192.168.0.168
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::11a4:cbff:feac:NaePh22
                                192.168.0.1

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

C:\Users\AmLuc>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-w T05]
           [-r count] [-s count] [[-j host-list] | [-k host-list]]
           [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
           [-q] [-s] target_name

Options:
```

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT:- ETHICAL HACKING

ping [google.com](https://www.google.com)

```
Command Prompt
192.168.0.1

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\Users\anuruping>

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-s TOS]
           [-r count] [-s count] [[-j host-list] | [-k host-list]]
           [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
           [-q] [-S] target_name

Options:
  -t           Ping the specified host until stopped.
               To see statistics and continue - type Control-Break;
               To stop - type Control-C.
  -a          Resolve addresses to hostnames.
  -n count    Number of echo requests to send.
  -l size     Send buffer size.
  -f          Set Don't Fragment flag in packet (IPv6-only).
  -i TTL      Time To Live.
  -s TOS      Type of Service (IPv6-only. This setting has been deprecated
               and has no effect on the type of service field in the IP
               header).
  -r count    Record route for count hops (IPv4-only).
  -s count    Timestamp for count hops (IPv4-only).
  -j host-list Loose source route along host-list (IPv4-only).
  -k host-list Strict source route along host-list (IPv4-only).
  -w timeout  Timeout in milliseconds to wait for each reply.
  -R          Use routing header to test reverse route also (IPv6-only).
               Per RFC 5095 the use of this routing header has been
               deprecated. Some systems may drop echo requests if
               this header is used.
  -S srcaddr  Source address to use.
  -c compartment Routing compartment identifier.
  -p          Ping a Hyper-V Network Virtualization provider address.
  -4          Force using IPv4.
  -6          Force using IPv6.
```

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT:- ETHICAL HACKING

netstat

```
Command Prompt
C:\Users\luc\cmdnetstat

Active Connections

Proto Local Address Foreign Address State
TCP 192.168.0.168:54583 localhost-192.168.0.168:54583 SYN_SENT
TCP 192.168.0.168:54582 localhost-192.168.0.168:54582 SYN_SENT
TCP 192.168.0.168:55128 localhost-192.168.0.168:55128 SYN_SENT
TCP 192.168.0.168:56008 localhost-192.168.0.168:56008 SYN_SENT
TCP 192.168.0.168:56009 20.196.146.34:https SYN_SENT
TCP 192.168.0.168:56018 4.186.223.158:https SYN_SENT
TCP 192.168.0.168:56011 a23-212-55-27:https SYN_SENT
TCP 192.168.0.168:56012 a23-212-55-24:https SYN_SENT
TCP 192.168.0.168:56013 a23-212-55-20:https SYN_SENT
TCP 192.168.0.168:56014 20.196.146.32:https SYN_SENT
TCP 192.168.0.168:56082 localhost-192.168.0.168:56082 SYN_SENT
TCP 192.168.0.168:62997 localhost-192.168.0.168:62997 SYN_SENT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:50015 [2801:4866:4042:408::223]:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:50173 localhost-192.168.0.168:50173 TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:50127 del11618-192.168.0.168:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:52362 localhost-192.168.0.168:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:52368 localhost-192.168.0.168:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:53182 [2801:4866:4042:408::223]:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:53529 google-192.168.0.168:https ESTABLISHED
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:53665 [2801:4866:4042:408::223]:https ESTABLISHED
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:53736 [2801:4866:4042:408::223]:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:53790 localhost-192.168.0.168:https ESTABLISHED
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:54160 [2801:4866:4042:408::223]:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:54361 [2801:4866:4042:408::223]:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:54775 localhost-192.168.0.168:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:55008 localhost-192.168.0.168:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:55111 localhost-192.168.0.168:https ESTABLISHED
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:55399 localhost-192.168.0.168:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:55885 localhost-192.168.0.168:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:56231 localhost-192.168.0.168:https ESTABLISHED
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:56381 [2801:4866:4042:408::223]:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:56678 [2801:4866:4042:408::223]:https ESTABLISHED
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:56681 [2801:4866:4042:408::223]:https ESTABLISHED
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:56888 google-192.168.0.168:https TIME_WAIT
TCP [2406:281:5::80fc:2d9d:60b:8fd:63de]:57363 del11618-192.168.0.168:https ESTABLISHED
```

4.traceroute google.com

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT:- ETHICAL HACKING

```
Command Prompt
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:56381 [2001:4860:4027:7900::]:https TIME_WAIT
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:56078 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:56081 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:56080 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:57783 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:57790 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:58044 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:58054 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:58389 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:58681 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:59038 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:60042 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:60035 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:60779 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:62082 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:62976 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:63638 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:63033 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:64027 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:64173 [2001:4860:4027:7900::]:https ESTABLISHED
TCP [2005:281:5:80fc:2d9d:62b:8fc:63de]:65069 [2001:4860:4027:7900::]:https ESTABLISHED

C:\Users\Amal>tracert www.google.com

Tracing route to www.google.com [2001:4860:4027:7900::]
over a maximum of 30 hops:

  0  0 ms  0 ms  0 ms  2005:281:5:80fc:1aff:cbff:feac:76e9
  1  *  *  *  Request timed out.
  2  *  *  *  Request timed out.
  3  27 ms  6 ms  6 ms  2005:283:408:180:172:30:1:22
  4  15 ms  6 ms  6 ms  2005:280:800:c80:1:2a0
  5  28 ms  7 ms  5 ms  2005:280:800:3168:61:4
  6  *  *  *  Request timed out.
  7  7 ms  6 ms  5 ms  2005:280:800:c80:1:2b0
  8  *  *  *  Request timed out.
  9  *  *  *  Request timed out.
 10  183 ms  10 ms  8 ms  2005:4860:4027:7900::
 11  35 ms  7 ms  6 ms  2005:4860:4027:7900::

Trace complete.

C:\Users\Amal>
```